Δ DELTA FILINGS

Corporate Governance Intelligence Platform

Functional Requirements Document

Version 3.0 — May 2026

Companion to: Delta Filings BRD v3

Table of Contents

1. Introduction

1.1 Purpose

This FRD decomposes the business requirements in BRD v3 into module-by-module functional specifications. Each feature block captures the purpose, actors, main flow, alternate flows and acceptance criteria required for implementation and verification.

1.2 Audience

- Engineering — implementation reference.
- QA — test-case derivation.
- Product — scope and acceptance alignment.
- Design — flow-by-flow UX contract.

1.3 Conventions

- Functional IDs follow the pattern FR-<module>-<sequence>.
- Priority uses MoSCoW (Must / Should / Could).
- 'System' refers to the Delta Filings Flask application running on Railway.
- All timestamps are IST unless stated otherwise.

1.4 Traceability to BRD

FRD Module	BRD Requirements
Dashboard (FR-D)	BR-001
Filings & Announcements (FR-A)	BR-002, BR-003, BR-004, BR-006, BR-007
Filing Diff (FR-FD)	BR-005
Draft Engine (FR-DR)	BR-010 — BR-018
DIN Search (FR-N)	BR-020, BR-021
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Price Alerts (FR-AL)	BR-042, BR-043, BR-044

FRD Module	BRD Requirements
Industry Intelligence (FR-I)	BR-045
Watchlist (FR-W)	BR-050, BR-051
Authentication & Onboarding (FR-U)	BR-052, BR-053, BR-056
Payments & Access (FR-PY)	BR-054, BR-055
Experience & Theme (FR-UX)	BR-060 — BR-064

2. Dashboard Module (FR-D)

FR-D-01 — Render the monthly filings calendar for the user's watchlist

Attribute	Value
ID	FR-D-01
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Give the user an at-a-glance view of all corporate filings across their watchlisted companies for a chosen month.
Preconditions	User is authenticated, has paid (or redeemed a referral) and has at least one watchlist entry.
Postconditions	User sees a calendar grid with per-day filing pills, plus recent activity, top tags and stats.

Main Flow

1. User navigates to /dashboard.
2. System loads the user's watchlist and resolves the active month (defaults to the current IST month).
3. System queries filings for those symbols within the month and groups them by date.
4. System computes stat tiles (filings tracked, this-week count, last-filing freshness) and recent activity (last 10 filings).
5. System renders the calendar, sector pulse card, and feature shortcuts.

Alternate / Error Flows

- Empty watchlist — system shows an onboarding empty-state with a CTA to add a company.
- Filings for a watchlisted company are missing or older than 6 hours — system kicks off an async backfill thread; UI shows a refreshing indicator on that company.

Acceptance Criteria

- Calendar reflects timezone-correct (IST) day boundaries for every filing.
- Recent activity row always shows up to 10 of the most recent filings across the watchlist.
- Async backfill does not block the dashboard render.

FR-D-02 — Sector pulse card with weekly AI briefing

Attribute	Value
ID	FR-D-02
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Give the user a one-paragraph briefing on what happened in their industry this past week.
Preconditions	User has chosen an industry during onboarding.
Postconditions	Card shows the briefing text plus the source-headline count tracked over the past 14 days.

Main Flow

6. Dashboard initiates a request to /api/industry-intel/{industry}.
7. System returns either the cached (≤ 30 min) result or recomputes it from Google News + curated profile + Gemini.
8. Card renders the briefing text and a 'Full briefing →' link to /industry/{industry}.

Alternate / Error Flows

- Industry not in the curated 16 — system shows a neutral fallback briefing with a 'limited coverage' note.
- Gemini unavailable — system falls back to a heuristic summary of the top headline.

Acceptance Criteria

- Briefing carries a visible 'machine-generated' label.
- The card is readable in both light and dark themes (dark-mode gradient override ensures legibility).

3. Filings and Announcements Module (FR-A)

FR-A-01 — Per-company filings list with pagination

Attribute	Value
ID	FR-A-01

Attribute	Value
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Let the user browse the full filing history for any tracked company.
Preconditions	Company symbol exists in the user's watchlist (or is known to the system).
Postconditions	User sees 15 filings per page, sorted newest-first, with tags and dates.

Main Flow

9. User opens /company/{symbol}.
10. System checks the local filings cache for the symbol.
11. If the cache is sparse (< 50 rows) or stale (> 6 h), system spawns an async backfill via fetch.py.
12. System paginates the available rows (15 per page) and renders them with tag badges.
13. User can click a filing to open the detail modal.

Alternate / Error Flows

- NSE returns 5xx — fetch.py falls back to a sample dataset; backfill error is logged.
- Concurrent backfill already in flight — in-process lock prevents duplicate scraping.

Acceptance Criteria

- First page renders even when backfill is still in progress.
- Tag pills use the same dark-mode tinted styling as the dashboard for visual consistency.

FR-A-02 — On-demand AI summary for a filing

Attribute	Value
ID	FR-A-02
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Reduce time-to-insight by summarising a filing on demand.
Preconditions	Filing exists; raw_text has been extracted (or can be) from its source PDF.
Postconditions	Summary is generated, persisted on the filings row, and shown in the UI.

Main Flow

14. User clicks 'Summarise' on a filing.
15. System ensures raw_text is present (extracts via pdf_parser.py if needed).
16. System calls Gemini (Flash for short text, Pro for long text) with a structured prompt.
17. System stores the summary back on the filings row and returns it.

Alternate / Error Flows

- Gemini API key missing — system returns a heuristic summary from the first ~200 characters of raw_text.
- Gemini quota exhausted — system surfaces a 'try again later' error and does not persist a stub.

Acceptance Criteria

- Summary carries a visible 'machine-generated' label.
- Summary persisted exactly once; subsequent views read from the database.
- Summary endpoint times out gracefully after 60 s.

FR-A-03 — Search and tag filter within a company

Attribute	Value
ID	FR-A-03
Priority	Should
Primary Actor(s)	Authenticated paid user
Purpose	Let the user narrow a company's filings by free text or category.
Preconditions	User is on /company/{symbol}.
Postconditions	Filings list re-renders to match the active filters.

Main Flow

18. User types a search query or clicks a tag chip.
19. System calls /api/search/{symbol} or /api/tags/{symbol}.
20. System returns matching filings; UI replaces the list, preserving pagination state.

Alternate / Error Flows

- No matches — UI shows an empty-state with a 'clear filters' CTA.

Acceptance Criteria

- Search responds within 500 ms at p95 on a populated cache.
- Tag chip filtering and free-text search are composable.

FR-A-04 — Extract and view embedded XBRL

Attribute	Value
ID	FR-A-04
Priority	Should
Primary Actor(s)	Authenticated paid user
Purpose	Surface structured XBRL data when the source PDF embeds it.
Preconditions	Filing has has_xbrl = true.
Postconditions	User sees parsed XBRL key-value pairs in a modal section.

Main Flow

21. User opens a filing detail modal.
22. System calls /api/xbrl/{id} which invokes pdf_parser.extract_xbrl().
23. UI renders a tagged keyed view (e.g., 'Net Profit', 'Date').

Alternate / Error Flows

- PDF has no embedded XBRL — endpoint returns 404; UI shows nothing in the XBRL section.
- Parser fails — endpoint returns the raw XML; UI shows a 'raw XBRL' fallback.

Acceptance Criteria

- XBRL extraction completes within 5 s at p95 for filings up to 5 MB.

4. Filing Diff Module (FR-FD)

FR-FD-01 — Auto-match and diff a filing against the previous similar filing

Attribute	Value
ID	FR-FD-01
Priority	Must

Attribute	Value
Primary Actor(s)	Authenticated paid user
Purpose	Let the user see what changed between successive filings of the same type.
Preconditions	Filing exists; at least one earlier filing of the same type and company exists in cache.
Postconditions	User sees inline and side-by-side diff views.

Main Flow

24. User clicks 'Compare' on a filing.
25. System calls /api/compare/{id} which invokes find_best_match() from database.py.
26. Match candidate selected by title similarity + date ordering.
27. diff_engine.compute_diff() runs over the two raw_text bodies.
28. diff_to_html() and side_by_side_html() return rendered HTML for the modal.

Alternate / Error Flows

- No reasonable match found — system returns 404 with a 'no prior filing to compare' message.
- Filings are too large — system truncates each side to ~200 KB before diffing.

Acceptance Criteria

- Diff result reflects line-level additions, deletions and information markers.
- Diff colours and dark-mode styling render correctly in both themes.

FR-FD-02 — Diff two arbitrary filings

Attribute	Value
ID	FR-FD-02
Priority	Should
Primary Actor(s)	Authenticated paid user
Purpose	Let the user explicitly compare any two filings of their choice.
Preconditions	Both filing IDs exist.
Postconditions	User sees the diff in inline and side-by-side modes.

Main Flow

29. User opens the filings picker, chooses two filings.
30. System calls /api/compare/{id_a}/{id_b}.
31. diff_engine renders the result; UI mounts it into the modal.

Alternate / Error Flows

- Filings have no raw_text — system extracts via pdf_parser before diffing.

Acceptance Criteria

- Diff renders within 8 s at p95 for two filings up to 100 KB each.

5. Draft Engine — SEBI Letter Wizard (FR-DR)

FR-DR-01 — Select filing type and pull past examples

Attribute	Value
ID	FR-DR-01
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Let the user pick a SEBI filing type and learn from past examples for the chosen company.
Preconditions	User is on /draft; watchlist contains the target company.
Postconditions	UI shows the filing-type registry and a list of past examples for the chosen type.

Main Flow

32. User opens /draft.
33. UI calls /api/draft/filing-types to load the registry of 20+ supported types.
34. User selects a type; UI calls /api/draft/examples for {company, type}.
35. System returns the most recent five matching filings; if none cached, system kicks off an async fetch.

Alternate / Error Flows

- No matches even after fetch — UI shows 'no past examples' and disables 'Learn style'.

Acceptance Criteria

- Filing-type registry covers all periodic and major event-based SEBI filings.
- Examples list reflects the watchlist company, not a global pool.

FR-DR-02 — Learn the company's writing style

Attribute	Value
ID	FR-DR-02
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Capture the tone and structure of a company's prior filings so the AI draft sounds right.
Preconditions	Past examples are available.
Postconditions	System returns a style fingerprint (tone, sentence length, formality markers).

Main Flow

36. User clicks 'Learn style'.
37. System sends the most recent past examples to Gemini with a structured prompt.
38. Gemini returns a structured fingerprint object.
39. UI persists the fingerprint into the draft session state.

Alternate / Error Flows

- Gemini unavailable — UI proceeds with a generic 'formal corporate tone' fallback.

Acceptance Criteria

- Fingerprint never includes personally identifying information.
- Subsequent drafts use the fingerprint without re-prompting until the user changes the filing type.

FR-DR-03 — AI-generate the draft body

Attribute	Value
ID	FR-DR-03
Priority	Must

Attribute	Value
Primary Actor(s)	Authenticated paid user
Purpose	Produce a first-draft SEBI letter combining the user's intent, the company's style and the chosen type's template.
Preconditions	Filing type selected; style fingerprint available (or skipped); user has filled the intent inputs.
Postconditions	System returns a structured letter (subject, body paragraphs, optional annexures).

Main Flow

40. User clicks 'Generate draft'.
41. Backend builds the prompt: type template + fingerprint + user intent + boilerplate from past examples.
42. Gemini returns the draft; system parses it into the editor schema.
43. UI mounts the result into the editor.

Alternate / Error Flows

- Gemini timeout — UI surfaces a retry CTA; no partial draft is committed.
- Output fails schema validation — system requests a single re-emission then surfaces a manual-fix CTA.

Acceptance Criteria

- Draft always includes a clearly labelled 'machine-generated' notice.
- Generation completes within 30 s at p95.

FR-DR-04 — AI-fill placeholders from public knowledge

Attribute	Value
ID	FR-DR-04
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Fill repetitive placeholders (CIN, registered office, ISIN) using public knowledge.

Attribute	Value
Preconditions	Draft contains at least one placeholder.
Postconditions	Placeholders are replaced with confident values; uncertain ones remain flagged.

Main Flow

44. User clicks 'AI-fill'.
45. System calls /api/draft/ai-fill with the placeholder list and target company.
46. Gemini returns values with confidence ratings.
47. UI updates the editor and highlights any low-confidence values for human review.

Alternate / Error Flows

- Gemini cannot find a value — placeholder kept as-is; UI marks it as 'manual-fill required'.

Acceptance Criteria

- Low-confidence fills are visually distinguished and excluded from export until reviewed.

FR-DR-05 — Improve, tone-match, formalise or custom-rewrite

Attribute	Value
ID	FR-DR-05
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Let the user iterate on the draft with one-click rewrites.
Preconditions	Draft has a non-empty body.
Postconditions	Body is replaced with the rewritten version; previous version remains in the editor history.

Main Flow

48. User clicks 'Polish' / 'Match tone' / 'Formal' / 'Custom'.
49. UI calls /api/draft/improve with the action and (if custom) the user's instruction.
50. Gemini returns the rewritten body; UI swaps it in.

Alternate / Error Flows

- User undoes — UI restores the previous body from history.

Acceptance Criteria

- Each improvement action is logged in the editor history with timestamp.

FR-DR-06 — Export draft as PDF and DOCX

Attribute	Value
ID	FR-DR-06
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Produce a print-ready deliverable.
Preconditions	Draft body is non-empty.
Postconditions	Browser downloads the chosen format.

Main Flow

51. User clicks 'Export PDF' or 'Export DOCX'.
52. Backend (draft_engine.py) renders the document via reportlab or python-docx.
53. Browser receives the file with the appropriate Content-Disposition.

Alternate / Error Flows

- Render fails — UI surfaces the error; no partial file is delivered.

Acceptance Criteria

- PDF and DOCX outputs preserve subject, body, signature block and annexure list.
- Export completes within 6 s at p95.

FR-DR-07 — Save, list and resume drafts

Attribute	Value
ID	FR-DR-07
Priority	Must
Primary Actor(s)	Authenticated paid user

Attribute	Value
Purpose	Persist drafts so the user can leave and return.
Preconditions	User is signed in.
Postconditions	Draft row is created (or updated) in user_drafts.

Main Flow

54. Editor autosaves on each meaningful change (debounced).
55. User can list previous drafts via /api/drafts and re-open one via /api/drafts/{id}.
56. User can delete via DELETE /api/drafts/{id}.

Alternate / Error Flows

- Save fails — UI surfaces 'unsaved changes' badge until retry succeeds.

Acceptance Criteria

- Drafts are strictly scoped to the owning user_id at the database query layer.
- Listing returns only the user's own drafts and the drafts they collaborate on.

FR-DR-08 — Team collaboration via invite links

Attribute	Value
ID	FR-DR-08
Priority	Should
Primary Actor(s)	Draft owner; invited collaborator
Purpose	Allow another user to view or edit a draft via a shareable token.
Preconditions	Draft exists and current user is the owner.
Postconditions	Token is generated; recipient gains the configured role on signin.

Main Flow

57. Owner clicks 'Invite' and chooses editor or viewer role.
58. System creates a row in draft_invites with a UUID token and expiry.
59. Owner shares the /draft/invite/{token} URL.

60. Recipient opens the URL; if signed in, system adds them to draft_teams with the configured role; if not, the token is stashed in the session and applied after signin.

Alternate / Error Flows

- Token expired or revoked — landing page shows a neutral 'invite no longer valid' message.
- Owner revokes a pending invite — token deleted; URL returns 404.

Acceptance Criteria

- Role changes and removals are owner-only and audited.
- Public landing page never reveals draft contents before claim.

6. DIN Search Module (FR-N)

FR-N-01 — Search directors across filings, governance reports and MCA history

Attribute	Value
ID	FR-N-01
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Return a comprehensive view of any director across all available sources.
Preconditions	User is on /din-search.
Postconditions	Results section lists matches grouped by source.

Main Flow

61. User submits a DIN or director name.
62. System validates input (DIN must be 8 digits; name must be ≥ 3 chars).
63. System runs search_din_across_companies() over filings text and governance XBRL.
64. In parallel, system calls mca_scraper to fetch director history (cached 30 days).
65. UI renders a unified results card with sources clearly labelled.

Alternate / Error Flows

- Invalid input — inline field-level error; no requests issued.
- MCA scrape blocked — system falls back to filings and XBRL only with a 'partial' badge.

Acceptance Criteria

- Cached responses (≤ 30 days) are reused without network calls.
- Results are ordered by recency where date is available.

7. Governance and XBRL Module (FR-G)

FR-G-01 — List a company's governance reports

Attribute	Value
ID	FR-G-01
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Surface the catalogue of governance disclosures filed through the NSE Integrated Filing system.
Preconditions	Symbol is known to the system.
Postconditions	UI shows a list of governance reports with type and date.

Main Flow

66. User opens the governance section on a company page.
67. System calls /api/governance/{symbol} which delegates to governance.list_reports().
68. List is rendered with select buttons for inspecting or comparing.

Alternate / Error Flows

- NSE returns no reports — UI shows an empty-state placeholder.

Acceptance Criteria

- List is cached server-side for 6 hours per symbol.

FR-G-02 — Inspect a single governance report (XBRL parse)

Attribute	Value
ID	FR-G-02
Priority	Must

Attribute	Value
Primary Actor(s)	Authenticated paid user
Purpose	Show the structured contents of a governance report.
Preconditions	Report ID is valid.
Postconditions	Parsed XBRL is rendered as a keyed list grouped by section.

Main Flow

69. User clicks a report row.
70. System calls /api/governance/{symbol}/{rec_id} which downloads the XBRL and runs the parser.
71. UI renders the parsed contents.

Alternate / Error Flows

- Parse fails — UI shows raw XML inside an expandable block.

Acceptance Criteria

- Parse completes within 8 s at p95.

FR-G-03 — Compare two governance reports

Attribute	Value
ID	FR-G-03
Priority	Should
Primary Actor(s)	Authenticated paid user
Purpose	Highlight changes between two governance reports of the same company.
Preconditions	Both report IDs are valid and belong to the same company.
Postconditions	UI shows a side-by-side compare view of the two parsed reports.

Main Flow

72. User selects two reports.
73. System calls /api/governance/compare/{symbol}/{rec_a}/{rec_b}.
74. Backend parses both and computes a structural diff.

75. UI renders the comparison.

Alternate / Error Flows

- One report fails to parse — UI shows it as 'unparseable' and renders the other side normally.

Acceptance Criteria

- Compare endpoint completes within 12 s at p95.

8. Compliance OS Module (FR-M)

FR-M-01 — Show obligations and scores for each watchlisted company

Attribute	Value
ID	FR-M-01
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Surface upcoming, overdue and completed regulatory obligations for the user's watchlist.
Preconditions	User is signed in and on /compliance.
Postconditions	Dashboard renders per-company score cards plus a unified obligations list.

Main Flow

76. User opens /compliance.
77. System calls /api/compliance which aggregates obligations + enforcement orders.
78. compliance.refresh_compliance is auto-triggered for any company that lacks recent data.
79. UI renders score cards, upcoming/overdue/completed groupings and the SEBI enforcement summary.

Alternate / Error Flows

- Watchlist is empty — UI prompts the user to add companies.

Acceptance Criteria

- All obligations link to their source statute or SEBI regulation (via sebi_link_resolver).

FR-M-02 — Manual refresh of compliance and enforcement data

Attribute	Value
ID	FR-M-02
Priority	Should
Primary Actor(s)	Authenticated paid user
Purpose	Let the user force a refresh when they suspect the cache is stale.
Preconditions	User is on /compliance.
Postconditions	Database rows for obligations / enforcement orders are updated.

Main Flow

80. User clicks 'Refresh' on a company card or globally.
81. System calls /api/compliance/refresh?symbol=X (or without symbol for all).
82. Refresh runs in a background thread; UI polls for completion.

Alternate / Error Flows

- SEBI portal unavailable — endpoint returns partial success; UI surfaces the affected company.

Acceptance Criteria

- Refresh never blocks the request thread for more than 1 s.

FR-M-03 — SEBI enforcement matching with confidence ratings

Attribute	Value
ID	FR-M-03
Priority	Must
Primary Actor(s)	System
Purpose	Map SEBI public orders to the companies a user tracks.
Preconditions	User has added at least one company.
Postconditions	enforcement_orders rows exist with a confidence rating (high / medium / low).

Main Flow

83. sebi_enforcement.refresh_enforcement_async() runs in the background.
84. For each company, scraper queries five SEBI order categories.
85. Title and party fields are normalised; a similarity score against the company's legal name yields the confidence rating.
86. Rows persisted with checked_at timestamp.

Alternate / Error Flows

- SEBI rate-limits — scraper backs off and resumes.

Acceptance Criteria

- At least 70% of stored orders carry a 'high' confidence rating against a labelled sample.

9. Stock Prices Module (FR-P)

FR-P-01 — Live prices, 3% movers and pre-open

Attribute	Value
ID	FR-P-01
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Surface live quotes for the watchlist, highlight movers and expose pre-open data.
Preconditions	User is on /prices.
Postconditions	UI shows live prices, a built-in 3% mover ribbon and pre-open alerts.

Main Flow

87. Page polls /api/prices on an interval.
88. stock_prices.py fetches NSE quotes for the watchlist plus the built-in 3% mover universe.
89. System applies any fired custom alerts and returns a unified payload.
90. Pre-open data is fetched separately on /api/pre-open.

Alternate / Error Flows

- NSE unavailable — endpoint returns cached values with a 'stale' badge.

Acceptance Criteria

- Watchlist tile updates within 60 s of an underlying NSE quote change.

FR-P-02 — Per-symbol quote with movement explanation

Attribute	Value
ID	FR-P-02
Priority	Should
Primary Actor(s)	Authenticated paid user
Purpose	Combine quote, recent filings and AI reasoning on /api/price/{symbol}.
Preconditions	Symbol is known.
Postconditions	UI renders the detailed card.

Main Flow

91. User clicks a symbol on the prices grid.
92. System calls /api/price/{symbol} for quote and recent filings.
93. User can request 'Deep intel' which calls /api/price-intel/{symbol}.
94. price_intel.py aggregates Google News, Reddit and recent filings, runs Gemini for a one-paragraph explanation, caches for 15 minutes.

Alternate / Error Flows

- Gemini unavailable — UI shows the raw inputs without the AI explanation.

Acceptance Criteria

- Cached intel returns in under 600 ms; fresh intel under 20 s at p95.

10. Price Alerts Module (FR-AL)

FR-AL-01 — Create, list, toggle and delete custom alerts

Attribute	Value
ID	FR-AL-01
Priority	Must

Attribute	Value
Primary Actor(s)	Authenticated paid user
Purpose	Let the user define their own price-movement alerts.
Preconditions	User is signed in.
Postconditions	price_alerts row created / updated / deleted.

Main Flow

95. User opens the alerts panel on /prices.
96. User fills symbol, threshold %, direction (up / down / either) and submits.
97. System upserts the alert via POST /api/alerts.
98. User can toggle, delete or test alerts inline.

Alternate / Error Flows

- Invalid symbol — inline error.
- Duplicate alert — system updates the existing row rather than creating a second.

Acceptance Criteria

- Each alert has a 60-minute cooldown enforced server-side.
- Alerts API never reveals other users' rows.

FR-AL-02 — Alert evaluation and Telegram delivery

Attribute	Value
ID	FR-AL-02
Priority	Must
Primary Actor(s)	System (alert_poller)
Purpose	Evaluate alerts against live prices and dispatch them.
Preconditions	Background poller has a lease for this cycle.
Postconditions	Eligible alerts fire (Telegram + in-browser); last_fired_at updated.

Main Flow

99. alert_poller.py runs every 60 seconds during IST market hours.
100. Worker acquires the cron_meta lease (CAS update); other workers no-op.
101. Worker queries enabled alerts whose cooldown has elapsed.
102. Worker fetches the latest prices via stock_prices and evaluates each alert.
103. Eligible alerts dispatched to Telegram (if linked) and surfaced to /api/prices.

Alternate / Error Flows

- Telegram dispatch fails — alert remains marked 'fired'; failure logged for the user to retry.

Acceptance Criteria

- Only one Gunicorn worker fires alerts per cycle.
- End-to-end delivery latency (threshold to Telegram) ≤ 90 s at p95.

FR-AL-03 — Link and unlink Telegram

Attribute	Value
ID	FR-AL-03
Priority	Should
Primary Actor(s)	Authenticated paid user
Purpose	Verify the user's Telegram identity and bind a chat_id to their account.
Preconditions	User has a Telegram account; Bot token configured.
Postconditions	user_telegram row created (or removed).

Main Flow

104. User clicks 'Link Telegram'.
105. Telegram Login Widget returns a signed payload to /api/alerts/link-telegram.
106. System verifies the HMAC signature against the Bot token.
107. Row persisted; alerts begin delivering to that chat_id.

Alternate / Error Flows

- Signature invalid — system refuses the link and shows an error.

Acceptance Criteria

- Chat IDs are never stored without a valid signature.

- Unlink purges the row immediately.

11. Industry Intelligence Module (FR-I)

FR-I-01 — Industry profile + live news + weekly briefing

Attribute	Value
ID	FR-I-01
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Give the user a curated and live view of their tracked industry.
Preconditions	User has chosen an industry (one of 16 curated).
Postconditions	UI shows the profile, four news buckets and the AI briefing.

Main Flow

108. User opens /industry or /industry/{industry}.
109. System calls /api/industry-intel/{industry}.
110. industry_intel.py returns a 30-min cached payload of: curated profile, four Google News RSS buckets (sector / M&A / policy / global), and a Gemini briefing.
111. UI renders the result with source attribution.

Alternate / Error Flows

- Industry not in the 16 curated — system serves a neutral fallback profile.

Acceptance Criteria

- Briefing is clearly labelled 'machine-generated'.
- News items link directly to the original sources.

12. Watchlist Module (FR-W)

FR-W-01 — Add and remove companies

Attribute	Value
ID	FR-W-01

Attribute	Value
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Let the user curate the companies they track.
Preconditions	User is signed in.
Postconditions	watchlist table updated; backfills triggered for added companies.

Main Flow

112. User adds a company via the sidebar add-company control.
113. System validates the symbol via /api/symbol_search if needed.
114. POST /api/watchlist inserts the row.
115. System spawns background backfill threads for filings + compliance + enforcement for the new company.

Alternate / Error Flows

- Duplicate symbol — request returns 200 with no-op (UNIQUE constraint).

Acceptance Criteria

- Backfill never blocks the watchlist API response.

FR-W-02 — Switch industry to replace watchlist

Attribute	Value
ID	FR-W-02
Priority	Must
Primary Actor(s)	Authenticated paid user
Purpose	Let the user reset their watchlist to the canonical set for a different industry.
Preconditions	User is signed in.
Postconditions	Watchlist replaced; backfills triggered.

Main Flow

116. User picks a new industry in settings.
117. POST /api/settings/industry calls replace_watchlist_with_industry which writes the new symbols atomically.
118. Backfills run in background for the new set.

Alternate / Error Flows

- Industry unknown — request returns 400.

Acceptance Criteria

- Watchlist update is atomic (single transaction).

13. Authentication and Onboarding (FR-U)

FR-U-01 — Sign up and sign in via Supabase

Attribute	Value
ID	FR-U-01
Priority	Must
Primary Actor(s)	Visitor
Purpose	Authenticate users without building a credential store.
Preconditions	Visitor has a valid email.
Postconditions	Supabase issues a JWT; Flask session is established.

Main Flow

119. Visitor opens /signup or /signin.
120. Flask form posts credentials to Supabase.
121. Supabase returns user metadata and a JWT.
122. Flask sets a server-side session keyed by user_id.

Alternate / Error Flows

- Invalid credentials — Flask flashes a neutral error.
- Supabase outage — login page shows a 'service unavailable' message.

Acceptance Criteria

- No raw passwords ever stored locally.
- Sign-in to dashboard round trip ≤ 2 s at p95.

FR-U-02 — Onboarding capture (industry, company, designation)

Attribute	Value
ID	FR-U-02
Priority	Must
Primary Actor(s)	New user
Purpose	Bootstrap the watchlist and personalise the dashboard for a new user.
Preconditions	User has just signed up and not completed onboarding.
Postconditions	Supabase user metadata updated; default watchlist seeded; user redirected to checkout.

Main Flow

123. User completes the /onboarding form.
124. System writes industry, company and designation into Supabase metadata.
125. System calls replace_watchlist_with_industry for the chosen industry.
126. Backfills queued; user redirected to /checkout.

Alternate / Error Flows

- User skips industry — system blocks submission with a field error.

Acceptance Criteria

- Onboarding can be completed in a single page submission.

14. Payments and Access (FR-PY)

FR-PY-01 — Razorpay order creation and signature verification

Attribute	Value
ID	FR-PY-01
Priority	Must

Attribute	Value
Primary Actor(s)	Authenticated user (post-onboarding)
Purpose	Charge the ₹4,999 annual access fee.
Preconditions	User is on /checkout and Razorpay is configured.
Postconditions	Verified payment is recorded; user is marked paid.

Main Flow

127. Frontend calls POST /api/payment/create-order; backend issues a Razorpay order for 499900 paise.
128. User completes the Razorpay checkout in the browser.
129. Frontend posts the payment_id + signature to /api/payment/verify.
130. Backend verifies the HMAC-SHA256 signature against the Razorpay secret.
131. On success, payments.mark_paid writes user_payments and updates Supabase metadata.

Alternate / Error Flows

- Signature mismatch — backend rejects with 400; nothing persisted.
- User cancels — checkout page remains; nothing persisted.

Acceptance Criteria

- User cannot be marked paid without a verified signature.
- Marking paid is idempotent on payment_id.

FR-PY-02 — Referral code redemption

Attribute	Value
ID	FR-PY-02
Priority	Must
Primary Actor(s)	Authenticated user (post-onboarding)
Purpose	Allow invited users to gain access without payment.
Preconditions	User has a valid referral code.

Attribute	Value
Postconditions	user_payments row written with method='referral'; user marked paid.

Main Flow

132. User types the code on /checkout.
133. POST /api/payment/redeem-referral validates the code and the user's eligibility.
134. On success, the system writes a referral redemption row and marks the user paid.

Alternate / Error Flows

- Code unknown or exhausted — UI shows a neutral error.
- User already redeemed — UI shows 'already redeemed' and proceeds.

Acceptance Criteria

- A single user cannot redeem the same code twice.

FR-PY-03 — Access gating across all paid endpoints

Attribute	Value
ID	FR-PY-03
Priority	Must
Primary Actor(s)	System
Purpose	Block all paid features for unpaid users.
Preconditions	User is signed in.
Postconditions	Unpaid users see /checkout; paid users proceed.

Main Flow

135. Flask wraps paid routes with require_paid decorator.
136. Decorator checks the user's paid flag (cached in session, falls back to Supabase admin API).
137. Unpaid users are redirected to /checkout.

Alternate / Error Flows

- Session cache stale — decorator re-reads from Supabase admin API.

Acceptance Criteria

- No /api/* paid route can be invoked without the paid flag.
- Logout always succeeds regardless of paid state.

15. Experience, Theme and Motion (FR-UX)

FR-UX-01 — Theme toggle with localStorage persistence

Attribute	Value
ID	FR-UX-01
Priority	Must
Primary Actor(s)	Any user
Purpose	Let the user switch between light and dark themes and have the choice remembered.
Preconditions	User loads any page with the theme toggle.
Postconditions	HTML root carries 'dark' class (or not); preference stored in localStorage.

Main Flow

138. Init script (in analytics.html) reads the stored preference and toggles the class on <html>.
139. User clicks the toggle in the page header.
140. Script flips the class and persists the new preference.

Alternate / Error Flows

- localStorage unavailable — preference defaults to light, toggle still works for the session.

Acceptance Criteria

- First paint matches the stored preference; no flash of incorrect theme.

FR-UX-02 — Dark theme palette and surface overrides

Attribute	Value
ID	FR-UX-02
Priority	Must

Attribute	Value
Primary Actor(s)	System
Purpose	Render every primary surface, card, badge, tag, modal, input and scrollbar legibly in dark mode.
Preconditions	html.dark is active.
Postconditions	All Tailwind utility classes (bg-white, bg-slate-*, text-gray-*, border-gray-*, ring-gray-*, divide-gray-*) are overridden to the dark palette.

Main Flow

141. styles.css declares the dark CSS variables and the html.dark utility overrides.
142. Tag and badge classes (.tag-*, .company-badge-*, .company-bg-*) have dedicated dark rules with tinted backgrounds and bright readable text.
143. Decorative gradients (e.g. Chemicals Pulse) carry dark overrides that preserve identity while remaining legible.
144. Modal, diff, scrollbar and spinner styles all carry dark variants.

Alternate / Error Flows

- A new template introduces an un-overridden utility — design review catches it; fix lands in styles.css.

Acceptance Criteria

- Every primary route passes a dark-mode contrast audit at AA on textual content.
- No element renders with a bright white background in dark mode.

FR-UX-03 — Animation pass — entrance, hover, theme transition

Attribute	Value
ID	FR-UX-03
Priority	Should
Primary Actor(s)	System
Purpose	Make the application feel smooth and intentional without being distracting.
Preconditions	User is on any page; prefers-reduced-motion not set.

Attribute	Value
Postconditions	Cards fade in and rise on load; hover states animate; theme toggle is smooth.

Main Flow

145. Cards apply the fadeInUp keyframe with a 0.4 s cubic-bezier easing on initial load.
146. Feature cards stagger their entrance by ~40 ms.
147. Cards, buttons, tags and nav items have 0.25 s colour and transform transitions.
148. Primary buttons gain a gradient + glow in dark mode and lift on hover.
149. Theme toggle icons rotate gently on hover.

Alternate / Error Flows

- User has prefers-reduced-motion: reduce — all animations collapse to ~0.01 ms via the media-query guard.

Acceptance Criteria

- No animation drops frames on a mid-tier laptop.
- Reduced-motion users see no motion.

FR-UX-04 — Inputs, focus ring and accessibility

Attribute	Value
ID	FR-UX-04
Priority	Must
Primary Actor(s)	Any user
Purpose	Keep interactive elements discoverable and keyboard-friendly in both themes.
Preconditions	User is on a page with form elements.
Postconditions	Inputs have visible focus rings; keyboard users can traverse all controls.

Main Flow

150. Inputs, textareas and selects use the dark surface with a 1px border in dark mode.
151. Focused fields get a 3px indigo ring at 18% opacity.

152. :focus-visible applies a 2px indigo outline to buttons and links.

Alternate / Error Flows

- User has high-contrast preferences — focus indicators retain visibility under forced colours.

Acceptance Criteria

- All primary flows can be completed with the keyboard alone.
- Focus indicator is visible against every supported surface colour.

16. Non-Functional Cross-Reference

Non-Functional Area	Applicable FRs
Performance	FR-D-01, FR-A-02, FR-DR-03, FR-DR-06, FR-G-03, FR-AL-02
Availability	FR-A-01, FR-AL-02, FR-M-02
Security	FR-U-01, FR-PY-01, FR-PY-03, FR-AL-03, FR-DR-07
Privacy	FR-U-01, FR-AL-03, FR-DR-07
Accessibility	FR-UX-01 — FR-UX-04 and all user-facing flows
Observability	FR-A-01, FR-AL-02, FR-M-03, FR-PY-01
Maintainability	FR-UX-02 (centralised theme overrides)